

INDIAN STATISTICAL INSTITUTE

Mathematics-3 (BSDS)

Endterm Examination

DATE: 10/12/2025

Total Marks: 50

Time: 10:30 am – 1:30 pm

MARKS WILL BE AWARDED ONLY FOR THOSE ANSWERS WITH PROPER JUSTIFICATION

Question 1: If $\{x_n\}_n$ is an orthonormal sequence in a Hilbert space $\mathcal{H}$, then show that

$$\lim_{n \rightarrow \infty} \langle x_n, y \rangle = 0,$$

for each $y \in \mathcal{H}$. [5]

Question 2: Assume that a sequence $\{x_n\}_n$ in an inner product space $\mathcal{H}$ satisfies $\langle x_n, x \rangle \rightarrow \|x\|^2$ and $\|x_n\| \rightarrow \|x\|$. Show that

$$x_n \rightarrow x \quad \text{in } \mathcal{H}.$$

[6]

Question 3: For two arbitrary vector subspaces M and N of a Hilbert space $\mathcal{H}$ establish the following:

$$(M + N)^\perp = M^\perp \cap N^\perp.$$

[6]

Question 4: Let $\{e_i\}_{i \in \mathbb{N}}$ be an orthonormal basis in a Hilbert space and let x be a unit vector, i.e., $\|x\| = 1$. Show that for each $k \in \mathbb{N}$ the set

$$\left\{ i \in \mathbb{N} : |\langle x, e_i \rangle| \geq \frac{1}{k} \right\}$$

has at most k^2 elements. [6]

Question 5: Let X be the set of all positive integers and $d(m, n) = |m^{-1} - n^{-1}|$. Show that (X, d) is not complete. [4]

Question 6: Let $\mathcal{T}$ be the collection of subsets U of $\mathbb{R}$ such that either $\mathbb{R} \setminus U = \emptyset$ or $\mathbb{R} \setminus U$ is infinite. Show that $\mathcal{T}$ is not a topology on $\mathbb{R}$. [5]

Question 7: Let $(X, \mathcal{T})$ be a topological space, and $A, B \subset X$ be connected subsets. Are the sets $A \cap B$ and $A \cup B$ always connected. Justify your answer. [6]

Question 8: Let (X, d) be a metric space. Define the distance of two non-empty subsets A and B of X by

$$d(A, B) := \inf\{d(x, y) : x \in A \text{ and } y \in B\}.$$

If $A \cap B = \emptyset$, A is closed, and B is compact, then show that $d(A, B) > 0$. [6]

Question 9: Let $f : (0, 1) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \sqrt{x} \log(x)$. Is f uniformly continuous. Justify your answer. [6]